

Phase 4 Capstone Project

Data Manipulation and Simulation

A step-by-step guide using R and Microsoft Excel

From a **project question** to **reproducible evidence** that your whole group can explain

What you will be able to do

Stage	Do	Show
Prepare	Organise, import, clean	Source log + clean file
Build	Define variables and rules	Formula + one worked row
Explore	Summarise before modelling	One table + one chart
Simulate	Generate uncertain outcomes	Distribution + risk measure
Challenge	Test assumptions and limits	Scenario table + checks
Defend	Explain decisions as a team	Evidence, caveats, judgement

Key takeaway

Choose the **smallest set of tools** that directly answers your **project question**.

The analysis story: six connected steps



Ask at every step

- ▶ What does one row represent?
- ▶ Which units and time period apply?
- ▶ Which inputs come from real data?

Check before moving on

- ▶ Can a teammate reproduce the result?
- ▶ Does the output answer the question?
- ▶ Are assumptions visible and testable?

Minimum evidence package

Component	Minimum work	What to show on slides
Question	Population, period, outcome, and decision	One precise investigation question
Real data	Source log, raw file, clean file, units, missingness	Source caption and small data summary
Variables	Transparent formula or rule	Formula and one worked row
Description	Centre, spread, range, missingness, chart	One table and one chart
Simulation	Repeated uncertain outcomes	Distribution, not one final number
Risk	Suitable downside quantity	Loss probability, percentile, VaR, ES, or shortfall
Sensitivity	Change at least two inputs	Compact scenario comparison
Verification	Manual or independent check	One “How we checked” box

Common mistake

Long code is not evidence. Simple, transparent work is stronger than an advanced model the group cannot explain.

First, organise your project files

Folder	Store	Key rule
01_raw_data	Original downloads	Never overwrite
02_clean_data	Approved clean tables	New file after cleaning
03_code	R scripts / formula notes	Re-runnable steps
04_output	Final tables and charts	Only presentation evidence
05_documentation	Source, units, assumptions, AI log	Keep an audit trail

Source record: five fields only

Organisation | [link](#) | download date | data period | definitions/units

Decide what one row means

Project type	One row could mean	Required identifier
Market	One fund on one trading day	Fund + date
Insurance	One policy for one year	Policy + year
Claims	One adverse event	Claim ID
Simulation	One trial or person-month	Trial/person + period

Complete before coding

Each row represents **one** _____ for **one** _____.

Common mistake

Do not mix units: daily, annual, policy and claim records need a clear structure.

A useful variable dictionary

Variable	Type	Meaning	Check
date	Date	Observation date or period	Valid and correctly ordered
id	Text	Anonymous row or scenario ID	Unique when intended
exposure	Numeric	Amount at risk, income, or value	Non-negative; unit stated
event	0/1	Whether the defined event occurred	Only 0 or 1
severity	Numeric	Loss before product rules	Missingness explained
payout	Numeric	Benefit after product rules	Between 0 and limit
return	Numeric	Change over a stated horizon	Decimal or percentage confirmed
scenario	Text	Named assumption set	Definition recorded once

Evidence to show

Include a short variable dictionary in your documentation, even if only selected fields appear in the final presentation.

Ready-to-use prompt: understand a dataset

Give AI	Ask AI to return
Question + one-row meaning	Likely meaning of each column
Column names + units	Type, missing, duplicate, range and consistency checks
Original definitions	Items that still need source confirmation
Tool: R or Excel	Cleaning steps in the correct order

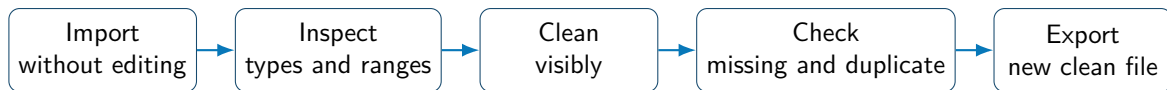
Copy-ready prompt

For [\[question\]](#), each row is [\[unit\]](#). Columns/units: [\[paste\]](#). Using [\[R/Excel\]](#), give a **short data dictionary, essential checks, source definitions to confirm**, and an **ordered cleaning plan**. **Do not invent values**; label uncertainty **“needs confirmation”**.

Evidence to show

Verify the answer against the **original source documentation**.

Workflow 1: import, inspect, clean, document



Key takeaway

Cleaning is not “making the data look nice”. It is a documented decision about errors, formats, missing values, and consistency.

A small practice dataset

Save as `demo_observations.csv`. It contains a missing value and inconsistent spelling on purpose.

```
date,id,category,exposure,event,severity
2026-01-01,A01,standard,12000,0,0
2026-01-02,A02,Standard,18500,1,2400
2026-01-03,A03,premium,30000,0,0
2026-01-04,A04,premium,27500,1,NA
2026-01-05,A05,standard,15000,1,900
2026-01-06,A06,standard,16000,0,0
```

Find before coding

- ▶ Which value is missing?
- ▶ Which category is inconsistent?

Do not assume

- ▶ Missing does not automatically mean zero
- ▶ Capitalisation should not create a new group

Clean the data in R

```
d <- read.csv("demo_observations.csv",
              stringsAsFactors = FALSE)
d$date <- as.Date(d$date)
d$category <- tolower(trimws(d$category))
stopifnot(!anyDuplicated(d$id))
stopifnot(all(d$event %in% c(0,1)))
stopifnot(all(d$exposure >= 0, na.rm=TRUE))
colSums(is.na(d))
d[!complete.cases(d), ]
write.csv(d, "demo_observations_clean.csv",
          row.names=FALSE)
```

Check	Expected result
Rows	6
Categories	standard, premium
Duplicate IDs	none
Invalid event	none
Negative exposure	none
Missing severity	1 row: A04
Output	new clean CSV

Clean the same data in Excel

Convert the range to an Excel Table with Ctrl+T; name it Data.

Purpose	Formula or action	Expected check
Standardise text	=LOWER(TRIM([@category]))	Only standard and premium
Flag missing	=IF(ISBLANK([@severity]), "Missing", "Observed")	Filter missing rows
Check duplicate ID	=COUNTIF(Data[id], [@id])	Unique IDs return 1
Check exposure	=IF([@exposure]>=0, "OK", "CHECK")	Every row returns OK
Validate event	Data Validation: whole number 0 to 1	No other values
Export	Copy approved columns to a new sheet	Raw sheet unchanged

Common mistake

Do not silently delete or replace missing values. Record the **decision** and its **effect**.

Ready-to-use prompt: generate cleaning steps

Tell AI	Require	Reject
Question, row unit, columns	Smallest workable steps	Unexplained mass deletion
Known data issues	Purpose + expected result	Missing = zero by default
R or Excel	A check after every step	Changes to the raw file

Copy-ready prompt

Clean [columns] for [question] in [R/Excel]. Known issues: [list]. **Keep raw data unchanged**. For each essential step, give **purpose**, **code/formula**, **expected result**, and a **pass/fail check**. Present options before changing missing values.

Common mistake

Use **anonymised samples only**; **do not upload restricted data**.

How should we treat missing data?

Choice	Use when	Must report
Recover from source	Value exists elsewhere	Retrieval rule
Keep missing	Analysis can use available cases	Affected calculations
Remove row	Exclusion is defensible	Number removed + bias risk
Impute	A stated model assumption is needed	Method + sensitivity result

Key takeaway

First ask: **why missing**, **how many rows**, **possible bias**, and whether another choice **changes the conclusion**.

Workflow 2: construct variables transparently

Project	Candidate variables	Evidence to show
Market	Return, loss, growth, drawdown	Formula, horizon, sign, unit
Portfolio	Weights, portfolio return, tail loss	Weight sum + worked row
Insurance	Event, severity, payout, premium, profit	Trigger + benefit rules
Life-related	l_x, d_x, q_x, p_x	Population + age interval

Evidence to show

For **one key variable**: **define symbols**, **state units**, **calculate one row**, and **test boundaries**.

Ready-to-use prompt: construct a variable

For returns, payouts, premiums, or other outcomes

I need to construct **[variable name]** for **[project purpose]**. Its rule is **[describe the rule in words]**. The relevant columns are **[names and units]**. Important assumptions are **[assumptions]**. I am using **[R / Excel]**.

Give me: (1) a mathematical definition with every symbol explained; (2) one worked example using simple numbers; (3) the shortest correct **[R code / Excel formula]**; and (4) checks for zero, missing, threshold, and maximum-limit cases. Keep assumptions in clearly named objects or cells rather than hiding numbers inside the formula.

Evidence to show

Manually reproduce the worked example and confirm that the software gives exactly the same answer.

$$R_t = P_t/P_{t-1} - 1, \quad r_t = \log(P_t/P_{t-1}), \quad L_t = -R_t$$

```
price$simple_return <- c(NA,  
  price$value[-1]/price$value[-nrow(price)]-1)  
price$log_return <- c(NA, diff(log(price$value)))  
price$loss <- -price$simple_return
```

Excel: =B3/B2-1 =LN(B3/B2) =-C3

Output	Expected result
First return	NA / blank
Price rises	Return > 0, loss < 0
Price falls	Return < 0, loss > 0
Units	Decimal or %, not mixed
Rows	Same as price table

A general payout rule

$$Y = I(\text{covered}) \min\{\max(L - d, 0)(1 - c), u\}$$

L : loss; d : deductible; c : co-payment; u : limit.

```
payout <- covered * pmin(  
  pmax(loss-deductible,0)*(1-copay), limit)
```

Excel: =A2*MIN(MAX(B2-\$H\$2,0)*(1-\$H\$3),\$H\$4)

Test row	Expected payout
Not covered	0
$L \leq d$	0
$d < L$	$(L - d)(1 - c)$
Severe loss	u at maximum
All rows	$0 \leq Y \leq u$

Workflow 3: describe what the data contain

Question	Statistic	Visual
How much data?	Valid n + missing count	None
What is typical?	Median + mean	Histogram
How variable?	SD + quartiles + range	Boxplot
Groups comparable?	Same summaries by group	Side-by-side plot
Outliers/skewness?	Mean vs median + tails	Histogram/boxplot

Key takeaway

Describe **first**; simulate only after the **data pattern is understood**.

Descriptive summaries in R and Excel

```
x <- na.omit(d$severity)
c(n=length(x), mean=mean(x),
  median=median(x), sd=sd(x),
  q25=quantile(x,.25), q75=quantile(x,.75),
  min=min(x), max=max(x))
```

Excel: COUNT, AVERAGE, MEDIAN, STDEV.S,
QUARTILE.INC, MIN, MAX

Demo output	Value
Valid <i>n</i>	5
Missing	1
Mean	660
Median	0
SD	1,028.6
Q1 / Q3	0 / 900
Min / Max	0 / 2,400

Expected values use the practice dataset after standardising category text; severity remains missing for A04.

Turn output into an interpretation

Sentence frame

"In the **observed sample** of _____ records, the **median** _____ was _____ units, while the **upper quartile** was _____. The **mean–median difference** and histogram suggest _____. **This matters because** _____."

Common mistake

Do not say only "the mean is 1,250". Explain what it represents and why it matters to the project decision.

Evidence to show

Every chart needs a descriptive title, labelled axes and units, a source note, and one insight sentence.

Ready-to-use prompt: summaries and charts

Ask for evidence, not decoration

My project asks **[question]**. I have cleaned data with **[sample size]** valid rows. The outcome variable is **[name, unit, and time period]**. Useful grouping or time variables are **[names]**. I am using **[R / Excel]**.

Recommend only the descriptive statistics and one or two charts that directly help answer the question. Provide the necessary **[R code / Excel formulas and chart steps]**, clear chart titles and axis labels, and a fill-in-the-blank interpretation sentence. Explain why each output is suitable. Do not claim causation or certainty, and do not create findings before seeing my actual output.

Key takeaway

After running the analysis, return the real output to AI only if permitted, then ask it to check whether your interpretation matches the numbers.

Workflow 4A: market and portfolio risk

Evidence	Define first	Answers
Return/growth	Horizon + simple/log	Performance path
Volatility	Window + frequency	Typical variation
Drawdown	Peak-to-trough rule	Historical decline
VaR / ES	Confidence + loss sign	Tail threshold/severity
Stress test	Shock rule + source	Recommendation robustness

Historical VaR and Expected Shortfall

$$\text{VaR}_q(L) = Q_q(L), \quad \text{ES}_q(L) = E[L \mid L \geq \text{VaR}_q(L)]$$

```
loss <- -portfolio_return; q <- 0.95  
VaR <- unname(quantile(loss,q))  
ES <- mean(loss[loss >= VaR])  
c(VaR_95=VaR, ES_95=ES)
```

```
Excel: =PERCENTILE.INC(Loss,0.95)  
       =AVERAGEIF(Loss,">="&VaRCell,Loss)
```

Output	Read as
VaR₉₅	95th loss percentile
ES₉₅	Mean loss at/above VaR
Expected relation	$\text{ES} \geq \text{VaR}$
Sign check	Larger positive = worse loss

Stress testing: ask “What if?”

Scenario	Asset A	Asset B	Interpretation
Ordinary adverse	−8%	−3%	Moderate joint loss
Macro shock	−12%	−16%	Diversification weakens
Liquidity stress	−25%	−10%	Tail loss dominated by Asset A

Common mistake

A stress test is not a prediction. Label scenarios as **historical**, **percentile-based**, or **hypothetical**.

Evidence to show

Explain the rule behind each shock and whether the recommendation survives it.

Workflow 4B: insurance or protection product

$$S = \sum_{j=1}^N Y_j$$

Part	Key input	Main check
Exposure	Policies / persons / periods	Unit and time basis
Frequency	Event probability or count rate	Data-supported range
Severity	Covered loss amount	Tail plausibility
Benefit rule	Trigger, deductible, co-pay, limit	Boundary cases
Output	Payout, profit, shortfall	Distribution, not one number

Key takeaway

Separate **real-data inputs** from **assumptions**, then **test the assumptions**.

Ready-to-use prompt: design a simulation

General simulation prompt

My OSALP project investigates **[question and decision]**. I want to simulate **[uncertain outcome]** using **[R / Excel]**. Real-data inputs are **[inputs and sources]**; assumptions without direct data are **[assumptions and ranges]**. One trial represents **[meaning of one trial]**, and I need **[profit / loss / payout / return / shortfall]** as the output.

Propose the simplest defensible simulation. State the algorithm in numbered plain-language steps before giving code or formulas. Separate data-supported inputs from assumptions. Include a reproducible random seed for R, or instructions to freeze final Excel values. Produce a distribution and **[selected risk measure]**. Add checks, at least two sensitivity tests, and limitations. Do not choose a probability distribution without explaining why it fits the variable and how I can assess that choice.

Common mistake

Simulation does not prove the future. Present results as **conditional on stated inputs and assumptions**.

A readable R simulation pattern

```
set.seed(20260908)
B <- 10000; m <- 1000; freq <- 0.04
simulate_year <- function(){
  nclaim <- rpois(1,m*freq)
  gross <- rlnorm(nclaim,meanlog,sdlog)
  paid <- pmin(pmax(gross-deductible,0)*
              (1-copay),limit)
  sum(paid)
}
claims <- replicate(B,simulate_year())
profit <- m*premium-claims-expense
c(mean_claims=mean(claims),
  loss_prob=mean(profit<0),
  median_profit=median(profit),
  adverse_profit=quantile(profit,.05))
```

Illustrative structure only; all product inputs need sources or labels as assumptions.

Output	Expected format
claims	10,000 totals
profit	10,000 outcomes
mean_claims	One currency value
loss_prob	Between 0 and 1
median_profit	50th percentile
adverse_profit	5th percentile

Excel simulation layout

Column	Example formula	Meaning
Trial	=ROW()-1	Simulation number
Claim count	=POISSON.INV(RAND(), ExpectedClaims)	Random count
Total severity	=MAX(0,NORM.INV(RAND(), MeanTotal,SDTotal))	Beginner approximation
Total payout	Product-rule calculation or lookup	Benefit after rules
Profit	=PremiumRevenue -TotalPayout-Expenses	Outcome per trial
Loss flag	=(Profit<0)	1 when loss occurs

Common mistake

RAND() changes whenever Excel recalculates. Freeze the final values used in the presentation and record the assumptions.

Life table thinking for age-dependent products

Column	Definition	Meaning	Check
l_x	alive at exact age x	survivor count	should not rise unexpectedly
d_x	$l_x - l_{x+1}$	deaths over one year	≥ 0
q_x	d_x / l_x	one-year death probability	between 0 and 1
p_x	$1 - q_x$	one-year survival probability	$p_x + q_x = 1$

Key takeaway

Use only when **mortality or survival directly affects cash flow**; state the **population** and **reference period**.

Compute life-table columns in R and Excel

```
life <- life[order(life$age),]  
life$dx <- c(head(life$lx,-1)-tail(life$lx,-1),NA)  
life$qx <- life$dx/life$lx  
life$px <- 1-life$qx  
stopifnot(all(life$qx>=0 & life$qx<=1,na.rm=TRUE))
```

Excel: $d_x = B2 - B3$ $q_x = C2/B2$ $p_x = 1 - D2$

Check	Expected result
Age order	Increasing
d_x	Non-negative
q_x, p_x	0 to 1
$q_x + p_x$	1
Final row	May be NA
Plot	q_x by age

Ready-to-use prompt: life-table application

Use only when survival or mortality affects cash flow

I have a life table for **[population and reference period]** with columns **[age, survivors, deaths, or probabilities]**. My product question is **[question]**. I am using **[R / Excel]**.

First, identify whether the table gives l_x , d_x , q_x , or p_x , without guessing. Then show how to derive only the missing columns needed for my project. Provide one manual age-interval calculation, the shortest code or formulas, and checks that probabilities stay between 0 and 1 and that survivor counts do not increase unexpectedly. Explain how the selected life-table quantity enters the product cash flow or simulation. Keep the source population and period visible.

Evidence to show

Check that the age definition and probability period match the proposed benefit and premium period.

Workflow 5: sensitivity and scenario analysis

Method	Change	Use it to
Sensitivity Scenario	One input at a time	Identify the main driver
	Coherent input set	Test an adverse environment
Examples Minimum	Frequency, severity, premium, limit Two separate inputs + one adverse scenario	Compare profit/risk/recommendation Show robustness

Build an assumption grid in R

```
grid <- expand.grid(  
  frequency=c(.02,.04,.07),  
  severity_mult=c(.8,1,1.4),  
  premium=c(80,95,110))  
grid$expected_loss <- m*grid$frequency*1800*  
  grid$severity_mult  
grid$revenue <- m*grid$premium  
grid$margin <- grid$revenue-grid$expected_loss  
grid <- grid[order(grid$margin),]
```

Check	Expected result
Grid rows	$3^3 = 27$
Frequency levels	3
Severity levels	3
Premium levels	3
Sort order	Worst margin first
Direction	Higher frequency lowers margin

Key takeaway

Screen simply first; fully simulate only the **scenarios that matter**.

Interpret sensitivity, do not only colour cells

Useful conclusion frame

“The **conclusion remained the same** when _____ varied from _____ to _____, but **changed** when _____ exceeded _____. Therefore, our recommendation is **most sensitive to** _____, which needs **monitoring or better data**.”

Evidence to show

Test at least two important assumptions separately, then show one coherent adverse scenario.

Five checks expected in every project

Check	Quick test	Pass evidence
Manual	Recalculate one row by hand	Exact match
Boundary	Zero, threshold, limit, missing	Expected behaviour
Unit	Currency, period, Direction	Raise one input
Output moves sensibly		
Reproducibility	Second member starts from raw data	Same key result

Key takeaway

Verification is evidence, not an optional final step.

Ready-to-use prompt: verify a completed analysis

Ask AI to challenge the work

Review the following analysis for **[project question]**: **[paste a concise description of data, assumptions, method, and actual output]**.

Act as a strict checker, not a co-author. Separate your response into: (1) calculation or coding risks; (2) data-definition and unit risks; (3) assumptions that need evidence; (4) missing boundary or sensitivity tests; and (5) claims that are stronger than the output supports. Do not invent new results. Finish with a short checklist that another group member can follow independently.

Evidence to show

A second member should complete the checklist and reproduce at least one key result from the raw data.

Automated checks and Excel audit tools

```
test_loss <- c(0,200,300,10000)
test_payout <- pmin(
  pmax(test_loss-deductible,0)*(1-copay),limit)
stopifnot(test_payout[1]==0)
stopifnot(test_payout[2]==0)
stopifnot(all(test_payout>=0))
stopifnot(all(test_payout<=limit))
```

Tool/check	Expected result
R stopifnot	No output if all pass
Zero loss	Payout 0
Below deductible	Payout 0
All payouts	0 to limit
Excel trace	Inputs visible
Reconciliation	premium – claims – expense = profit

Ready-to-use prompt: debug without blind copying

Paste the exact error and a minimal example

I am using **[R / Excel]**. My intended result is **[describe it]**. The exact error or unexpected output is **[paste it]**. The smallest relevant code, formula, and sample data are **[paste them]**.

Diagnose the likely cause using only the information provided. Explain the cause in student-friendly language, show the smallest correction, and state what changed. Then give one test that would fail before the correction and pass afterwards. If information is insufficient, ask me to inspect a specific object, cell, data type, or intermediate value rather than rewriting the whole analysis.

Key takeaway

Keep a short debugging record: original problem, correction, reason, and verification result.

Use AI as an assistant, not as evidence

Stage	Acceptable assistance	Human verification
Data search	Suggest sources and keywords	Open original source; confirm definitions and units
Code draft	Generate a small function or formula	Test a manual row and boundaries
Debugging	Explain an error message	Record the final change and why it is correct
Interpretation	Suggest questions	Rewrite against actual output
Presentation	Improve structure and captions	Keep evidence, citation, and limitations visible

Common mistake

Never present invented “realistic” data as real, and never retain code that the group cannot explain.

Suggested AI-use log

Date and tool	Purpose of prompt	Output used?	Human verification and revision
		Yes / No / Partly	
		Yes / No / Partly	
		Yes / No / Partly	

Key takeaway

The group remains responsible for the source, method, code, interpretation, and final claim.

Turn the analysis into six presentation slides

#	Slide	Show	Say
1	Question	Population, horizon, decision	Why it matters
2	Data	Source, sample, units, limits	What one row means
3	Method	Formula / simulation logic	Why this method
4	Evidence	One table + one chart	Main pattern
5	Robustness	Distribution + scenarios	What changes conclusion
6	Judgement	Conditional recommendation	Limit + next step

Common mistake

No worksheet screenshots. Extract only the **evidence needed for the message**.

Ready-to-use prompt: convert results into slides

Create a concise evidence story

Our project question is **[question]**. Our data, method, actual findings, sensitivity results, and limitations are **[paste concise notes]**.

Organise them into six slides: question and decision; data and definitions; method; main evidence; simulation and sensitivity; judgement. For each slide, provide one headline, no more than three short bullets, the best table or chart to show, and one speaker explanation. Preserve all qualifications and source notes. Do not add findings, statistics, or causal claims that are not in my notes. End with a conditional recommendation and one practical next step.

Key takeaway

Use the generated outline as a draft. Your group must select, verify, and explain every retained statement.

Submission-ready checklist

- ☐ Precise, feasible question
- ☐ Raw data unchanged
- ☐ Source log and units complete
- ☐ Missingness and duplicates checked
- ☐ Formula and manual example shown
- ☐ Descriptive evidence before simulation
- ☐ Inputs supported or labelled
- ☐ Simulation result is stable enough
- ☐ Two key assumptions tested
- ☐ Independent result check done
- ☐ AI assistance recorded
- ☐ Conclusion has conditions and limits

Key takeaway

Every member should be able to **explain** the **question, data, simulation, findings, and recommendation**.

Your next working session

Step	Build together	Cross-check owner
1	One-sentence question + one-row meaning	All members
2	Folder structure + source log	Data lead
3	Clean a small sample	Second member verifies
4	Key variable + worked row	Method lead
5	One summary table + chart	Presentation lead
6	Simulation logic + one sensitivity test	Another member reproduces

Key takeaway

Finish fewer **outputs**, but make every output **explainable and reproducible**.

Question → Evidence → Judgement

The best model is one your whole group can **explain**, **verify**, and **defend**.

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Retain data sources, assumptions, formulas, code, AI-use records, checks, and presentation outputs.
